Convex and Quasi-convex Functions

Convex Optimization, Take-home Quiz 3

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English summary of my original Persian report, which is kept as a PDF in this folder.

The original is a scanned handwritten answer sheet in Persian (`report_fa.pdf`).

The quiz had nine questions on convex, concave and quasi-convex functions. My answers:

1. **Sublevel sets.** A function is quasi-convex exactly when all its sublevel sets are convex; I proved both directions. Example: $f(x) = -x^2$ on $x \geq 0$ is not convex, but its sublevel sets are intervals.
2. **Rank of PSD matrices.** $\operatorname{rank}(X)$ on $\mathbf{S}_+^n$ is not quasi-convex. The counterexample is $X = \operatorname{diag}(1,0)$, $Y = \operatorname{diag}(0,1)$, $\theta = \frac{1}{2}$. It is quasi-concave, which I proved through $\ker(X + Y) = \ker X \cap \ker Y$ for $X, Y \succeq 0$.
3. **A convex function and an inequality.** $x/(1-x)$ is convex on $[0,1)$, since $f'' = 2/(1-x)^3 > 0$. Applying Jensen's inequality to it gives $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$.
4. **Three functions.**
   - The logistic function $1/(1+e^{-2x})$ is convex for $x < 0$ and concave for $x > 0$.
   - $n \operatorname{tr}(X) \log \operatorname{tr}(X) - \operatorname{tr}(X) \log\det X$ is convex on $\mathbf{S}_{++}^n$. The proof works with $t = \operatorname{tr} X$ and a perspective.
   - $\binom{x}{r}$ is convex on $(r-1, \infty)$. I showed this through $\log\binom{x}{r} = \sum_k \log(x-k) - \log r!$.
5. **Quasi-concavity on an interval.** A continuous f on $[a,b]$ is quasi-concave if $f(\theta a + (1-\theta)b) \geq \min\{f(a), f(b)\}$.
6. **Separating functions.** A convex f and a concave g with $g \leq f$ can be separated by an affine function h with $g \leq h \leq f$.
7. **Determinant root.** $(\det X)^{1/n}$ is concave on $\mathbf{S}_{++}^n$. I restricted it to a line $Z + tV$, which reduces it to a geometric mean of $1 + t\lambda_i$.
8. **A perspective.** $\|x\|_p^p / t^{p-1}$ $(p > 1)$ is convex as the perspective of the convex function $\|x\|_p^p$.
9. **Square of a function.** f^2 is convex when f is convex and non-negative, or concave and non-positive. The proof uses $(f^2)'' = 2f'^2 + 2ff''$.